BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, BANNU

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2023

BIOLOGY

Past Paper 2023 • Paper Code: BN-IX-XXIII-I

Retyped faithfully from the original board paper — same questions, no changes.

SECTION – A (Version-B)

Time Allowed: 20 Min | **Maximum Marks:** 12 | **Paper Code:** BN-IX-XXIII-I

Note: There are THREE sections in this paper i.e. A, B & C. Attempt section A and return it to the superintendent within the given time. You have four choices for each objective type question as A, B, C and D. The choice which you think is correct fill that circle in front of the question number. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero mark in that question.

Q1. Which one of the following statement is correct regarding anaphase-II?

(A) Chromosomes arrange themselves on the equatorial plate

(B) Sister chromatids of each chromosomes move to the opposite pole

(C) Sister chromatids of each chromosomes reach to the opposite poles

(D) Centrioles move to opposite poles and make spindle fibres

Q2. During temporary enzyme-substrate complex, which one of the following is converted to product?

(A) Prosthetic groups

(B) Enzyme

(C) Active site

(D) Substrate

Q3. The process in which CO_2 is reduced to form glucose is:

(A) Carbon cycle

(B) Calvin cycle

(C) Nitrogen cycle

(D) Kreb cycle

Q4. To carry out the normal process of photosynthesis plants require:

(A) High temperature

(B) Low temperature

(C) Optimum temperature

(D) No effect of temperature

Q5. When diarrhea is due to bacterial infection, it is treated by:

(A) Taking of alkaline medicines
(B) Taking of juices
(C) Taking of antibiotics
(D) Taking of dietary fiber

Q6. Which of the following is incorrect about human heart:

(A) Protected by pericardium
(B) Has four chambers
(C) Has thin walled atria
(D) Left ventricle of heart is smallest chamber

Q7. Stem: "Immunology" is the study of:

(A) Defence system of body
(B) Welfare humanity
(C) Drugs and their effects
(D) Delivery of drugs

Q8. In scientific method, initially hypotheses are developed through certain.

(A) Experiments
(B) Conclusions
(C) Observations
(D) Deductions

Q9. Penicillium belongs to which of the following group of organisms?

(A) Algae
(B) Plants
(C) Fungi
(D) Animals

Q10. Red cell easily carry more oxygen, because they are:

(A) Thin and long in size
(B) Disc like and biconcave in shape
(C) Thick and long in size
(D) Tube like and biconcave in shape

Q11. Which of the following pair is an example of prokaryotic cells?

(A) Plants and bacteria
(B) Animals and cyanobacteria
(C) Bacteria and cyanobacteria
(D) Protista and Fungi

Q12. The cell division in which diploid chromosomes are reduced to haploid number is:

(A) Mitosis
(B) Meiosis-I
(C) Meiosis-II
(D) Mitosis and Meiosis-II

SECTION – B

Time Allowed: 2:40 Hrs | **Maximum Marks:** 32 | **Paper Code:** BN-IX-XXIII-I

> Note: Attempt section B & C accordingly.

Q2. (Marks 32) Q.No 2. Attempt any EIGHT parts of the following. All parts carry equal marks.

i. Briefly explain the importance of horticulture as a profession.

ii. Briefly describe relationship of virus with its host.

iii. Read the following statement carefully and write the name of the scientists.

- Statement | Scientists
- All living organisms are made of tissues
- Plants are made of cells
- First time observed cork cell under microscope
- Every cell comes from a cell

iv. Write two functions of xylem and phloem in plants.

v. Define karyokinesis also write names of its main phases.

vi. With the help of diagrams show differences between early anaphase and late anaphase.

vii. Explain lock and key model of enzyme action with the help of diagram.

viii. Differentiate between lactic acid fermentation and alcoholic fermentation.

ix. Column A contains some properties of vitamin D. write their answer in column B.

- Column A | Column B
- Group
- Main function
- Signs of deficiency
- Main sources

x. Define guard cells and their role in the opening and closing of stomata.

xi. Differentiate between the role of bicuspid valve and semi lunar valve in blood circulation.

SECTION – C

Maximum Marks: 21 | **Paper Code:** BN-IX-XXIII-I

> Note: Attempt any THREE of the following. All questions carry equal marks

Q3.

a) Briefly explain how scientists use data in scientific method.

b) Enlist any four steps taken for the conservation of biodiversity of Pakistan.

Q4.

a) Why mitochondria is called power house of eukaryotic cell?

b) Write formation of spindle fibre in plant cells.

Q5.

a) Differentiate between coenzymes and activators.

b) Write steps involved in electron transport chain.

Q6.

a) Write three examples of each of organic fertilizers and inorganic fertilizers.

b) Differentiate among aorta, arterioles, capillaries and venules.